Data-Driven Inventory Planning for a Retail Pharmacy

A Mid-Term report for the BDM capstone Project

Submitted by

Name: SPANDAN BHOI

Rollnumber: 23f3002227

IITM Online BS Degree Program,

Indian Institute of Technology, Madras, Chennai

Tamil Nadu, India, 600036

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1.Executive Summary

Rahul Medical Store is a local pharmacy in Sambalpur, Odisha, operating since 2008. Over the years it has grown steadily, but while working with the owner I found that there are some recurring issues such as expired stock, difficulty in predicting demand, and problems arising from customers asking for medicines without proper prescriptions. At present, inventory is mainly tracked in Excel along with staff supervision, which is useful but not sufficient for long-term planning.

Through this project, my aim is to help the store improve its inventory management practices. I have collected primary data from the shop in the form of sales registers, purchase records, and expiry reports. I used Excel pivot tables to study sales trends, ABC analysis to identify important products, and expiry checks to understand where losses may happen. These tools were chosen because they are simple and can be applied directly by the owner or staff without technical difficulty.

From the preliminary analysis, I observed that a small group of medicines (Category A items) contribute to most of the sales, while a large number of slow-moving medicines contribute very little and are at higher risk of expiry. If the shop adopts structured tracking and monitoring, it can reduce expiry losses, improve stock planning, and handle non-prescription sales more responsibly. This midterm submission covers the initial findings, and I will refine the analysis further in the final stage with more forecasting and recommendations.

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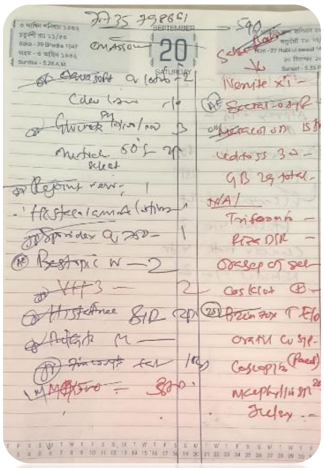
2. Proof of Originality of the Data

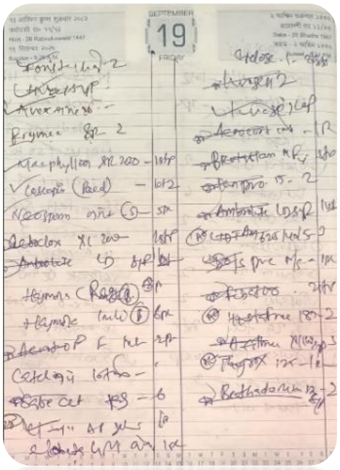
- Business Name: Rahul Medical Store
- Address: Budharaja Road, Near Varun Plaza, Sambalpur, Odisha
- Owner's Name: Mr. Sanjay Kumar Pradhan

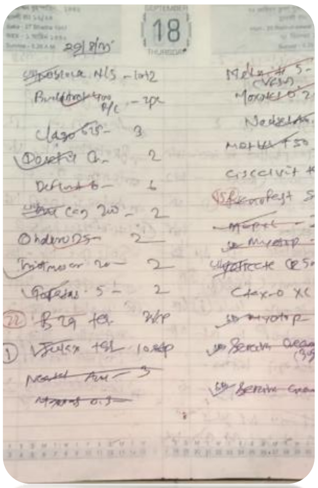
Data Collected: Sales and Purchase Register (April–June 2024)
Video of Interaction with Business Owner: VIDEO
Letter of Permission / Confirmation: NOC

Google Drive Link (Excel Data): DATA

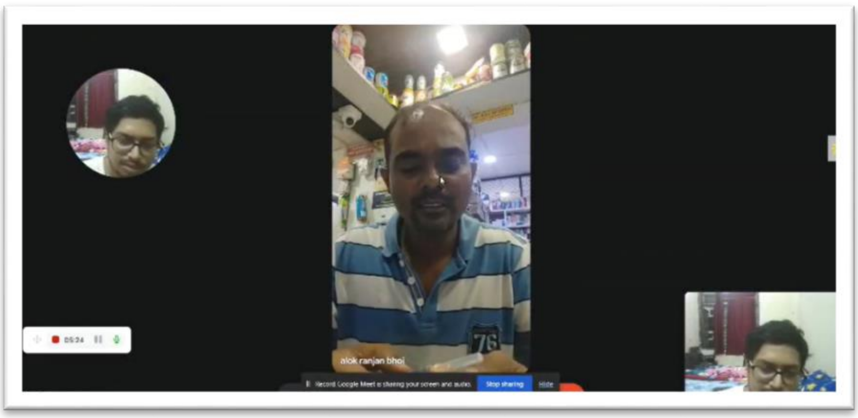
SHOP BOARD







PRIMARY DATAS



INTERACTION WITH OWNER

3.Metadata

The dataset for Rahul Medical Store has been compiled into an Excel workbook. It contains both sales and purchase records for the period April 2025 – June 2025, organized into different sheets. Each worksheet and its fields are described below:

1. Sales Register

This sheet records **all medicines sold** during the study period.

Period:- April 25 - June 25								
Date	Product Name	Batch No	Expiry Date	Quantity	MRP (₹)	GST (%)	GST Amount (₹)	Amount (₹)

Columns:

- Date – Date of sale.
- Customer Name – Customer or institution (if recorded).
- Product Name – Name of the medicine sold.
- Batch Number – Batch reference of the medicine sold.
- Expiry Date – Expiry of the sold batch.
- Quantity – Number of units sold.
- Rate (₹) – Selling price per unit.
- Amount (₹) – Quantity × Rate.
- GST % – GST rate applicable.
- GST Amount (₹) – Tax component of the sale.
- Total (₹) – Amount + GST.

2. Purchase Register

Period:- April 25 - June25										
Date	Customer/Supplier	Product Name	Batch No	Expiry Date	Quantity	Rate (₹)	Amount (₹)	GST (%)	GST Amount (₹)	Total (₹)

- This sheet records all medicines purchased from suppliers.
- Columns:

- Date – Date of purchase.
- Customer/Supplier – Name of supplier/distributor.
- Product Name – Name of medicine procured.
- Batch Number – Batch reference for tracking.
- Expiry Date – Expiry of purchased stock.
- Quantity – Number of units purchased.
- Rate (₹) – Purchase price per unit.
- Amount (₹) – Quantity × Rate.
- GST % – GST rate applicable.
- GST Amount (₹) – Tax paid.
- Total (₹) – Amount + GST.

3. Monthly Sales Sheets (Sales April, Sales May, Sales June)

Date	Product Name	Batch No	Expiry Date	Quantity	MRP (₹)	GST (%)	GST Amount (₹)	Amount (₹)

- These are filtered views of the Sales Register, showing month-wise sales.
- Columns are same as Sales Register, but restricted to the respective month.
- Purpose: Makes it easier to analyze monthly trends in sales.

4. Monthly Purchase Sheets (Pur April, Pur May, Pur June)

Date	Customer/Supplier	Product Name	Batch No	Expiry Date	Quantity	Rate (₹)	Amount (₹)	GST (%)	GST Amount (₹)	Total (₹)

- These are filtered views of the Purchase Register, showing month-wise procurement.
- Columns are same as Purchase Register, but restricted to the respective month.
- Purpose: Supports comparison of purchases vs. sales on a month-by-month basis.

5. Derived / Analytical Fields

- To support insights, additional computed fields will be generated:
- Revenue (₹) = Quantity × Rate (for sales).
- Cost of Goods Sold (COGS, ₹) = Quantity × Purchase Rate (from Purchase Register)..

- Expiry Risk Flag = medicines expiring within next 3 months.
- ABC Category = classification of medicines based on revenue contribution.

LINK TO THE PROJECT DATA: BDM DATASET

4.Descriptive Analysis

1. Overview of Sales & Purchases

Metric	Value
Unique Products Sold (SKUs)	51
Total Sales (₹)	6,57,774.34
Total Purchases (₹)	4,94,586.90
Average Monthly Sales (₹)	2,19,258.11
Average Quantity per Product	6.18 units

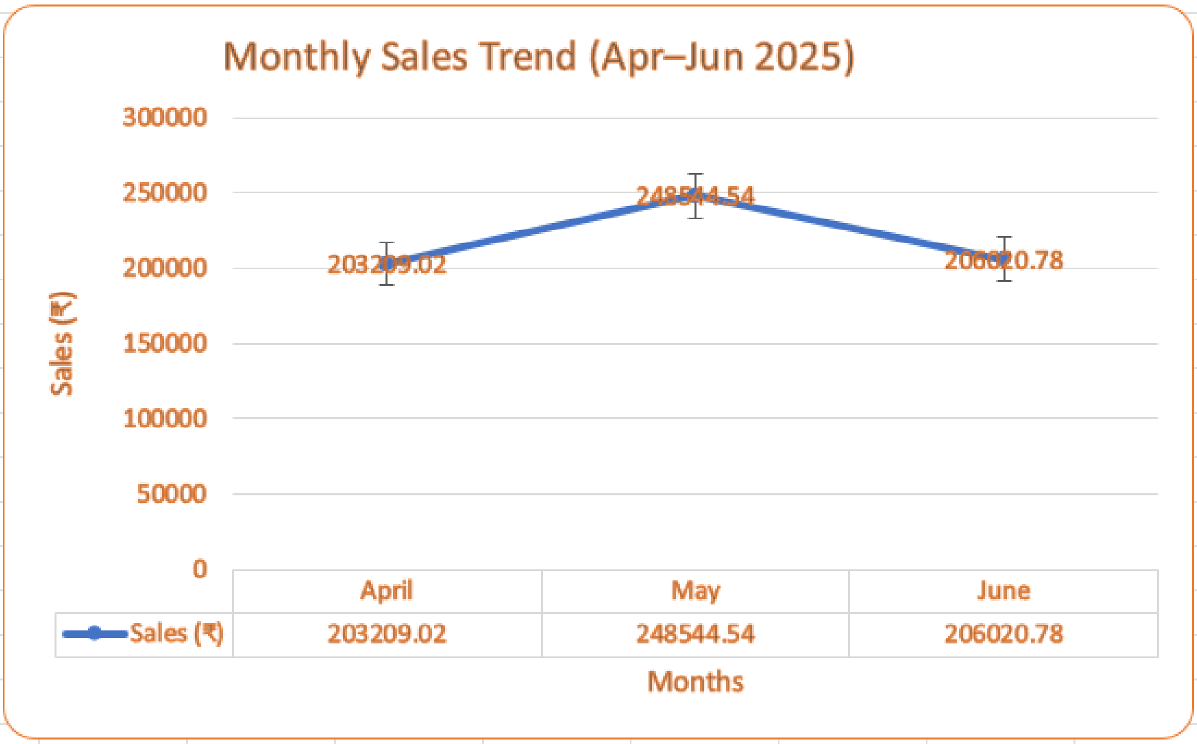
Sales exceeded purchases, which indicates healthy turnover. However, some products may still be overstocked or slow-moving, requiring deeper analysis.

2. Monthly Trends

Month	Sales (₹)	Growth vs Previous Month
April	**2,03,209.02**	**-**
May	**2,48,544.54**	**+22.3%**
June	**2,06,020.78**	**-17.1%**

- Sales peaked in May (₹2.48 lakh), a 22% increase over April, possibly due to seasonal demand.

- June saw a 17% decline compared to May, indicating demand fluctuations.
- Overall trend shows a mid-quarter peak, suggesting the need for better inventory planning.

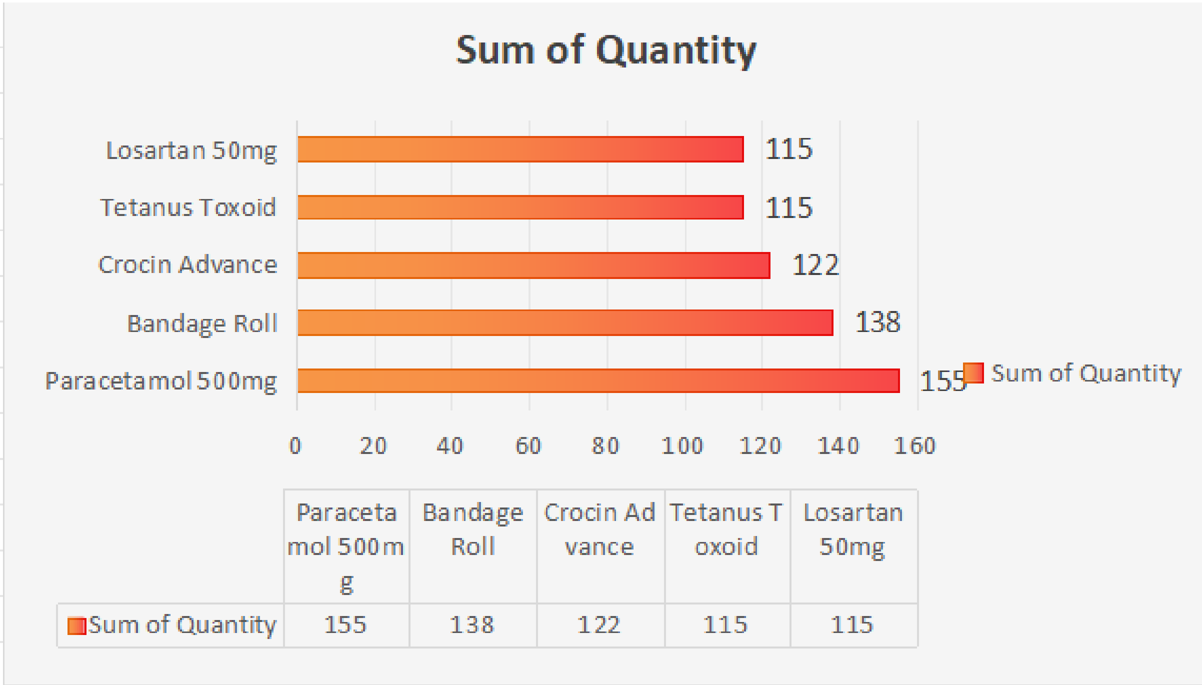


3.Product-Level Analysis

Product Name	Sum of Quantity	Sum of Amount (₹)
Paracetamol 500mg	155	1562.4
Bandage Roll	138	3116
Crocin Advance	122	2459.52
Tetanus Toxoid	115	1305.25
Losartan 50mg	115	11103.25

- Paracetamol 500mg was the highest-selling item with 155 units sold, followed closely by Bandage Roll (138 units) and Crocin Advance (122 units). This indicates a strong and consistent demand for basic medical supplies.
- From a revenue perspective, certain products demonstrated high value. For instance,

- Losartan 50mg generated significant revenue despite selling fewer units compared to top-volume items, highlighting its higher price point and strong contribution to overall profitability
- High-risk items: Multivitamin Syrup, Nicotine Patch, Thermometer – each contributes ₹1,500–₹3,000+ in expiry risk.
- Repeated small risks: Paracetamol 500mg, Prednisolone 10mg – low value individually, but frequent.
- Pareto effect: 20% of products drive 80% of Value at Risk.
- Key suppliers linked to risks: Popular Medical Store, Taj Pharmasutical.
- Actions: prioritize clearance of slow movers (discounts, bundling), negotiate for longer shelf-life stock, improve demand forecasting.



4.Inventory Management Analysis

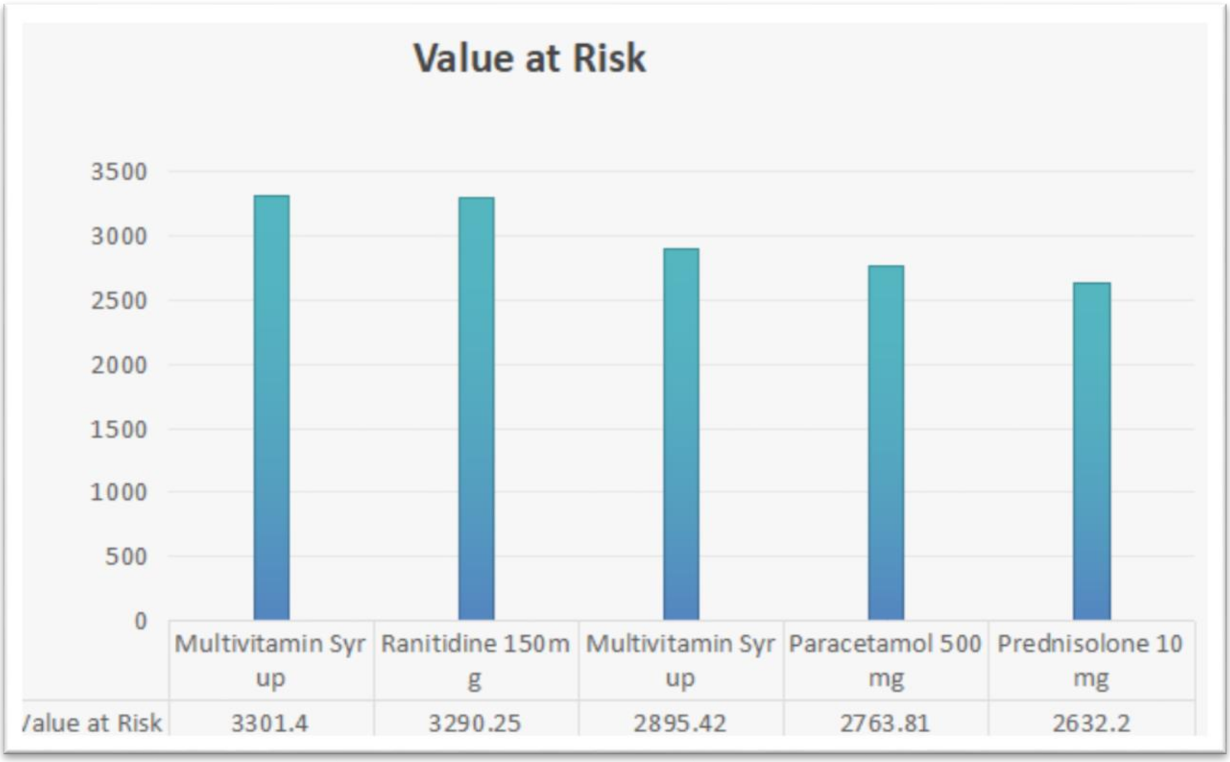
A comparison of purchased quantities versus sold quantities reveals critical inventory imbalances. Several essential medicines experienced **stockouts**, where demand outstripped the available supply. While preventing stockouts is important, excess inventory increases holding costs and the risk of product expiry.

Expiry Risk Analysis

The analysis identified several products with a near-term expiry date, posing a financial risk to the store. The total "Value at Risk" from these items is significant,

Date	Customer/Supplier	Product Name	Expiry Date	Safety	Value at Risk
2025-04-01	Popular Medical Store	Multivitamin Syrup	Dec-25	At Risk	3301.4
2025-04-04	Popular Medical Store	Ranitidine 150mg	Dec-25	At Risk	3290.25
2025-04-04	Popular Medical Store	Multivitamin Syrup	Dec-25	At Risk	2895.42
2025-04-05	Popular Medical Store	Paracetamol 500mg	Dec-25	At Risk	2763.81
2025-04-06	Taj Pharmasutical	Prednisolone 10mg	Dec-25	At Risk	2632.2

- Multivitamin Syrup and Ranitidine 150mg each contributing over ₹3,000 to this risk.
- Popular Medical Store and Taj Pharmasutical, are predominantly linked to the stock at risk of expiry.



Closing Stock Analysis

The closing stock analysis reveals critical inventory management challenges that directly impact the store's profitability and customer service levels.

Product Name	Purchased Qty	Sold Qty	Closing Stock
Amoxicillin 250mg	25	63	-38
ORS Sachet	55	86	-31
Dolo 650mg	61	91	-30
Erythromycin 250mg	55	84	-29
Disprin	78	104	-26
Atorvastatin 10mg	77	77	0
Becosules	80	80	0
Ibuprofen 400mg	60	60	0
Tetanus Toxoid	115	115	0
Thermometer	70	70	0
Calpol 500mg	61	48	13
Domperidone 10mg	85	72	13
Metformin 500mg	65	51	14

Montelukast 10mg	100	86	14
Avil 25mg	130	114	16
Combiflam	118	98	20

Negative Closing Stock (Stockouts): The negative values indicate demand exceeded available inventory, resulting in lost sales opportunities. Key findings:

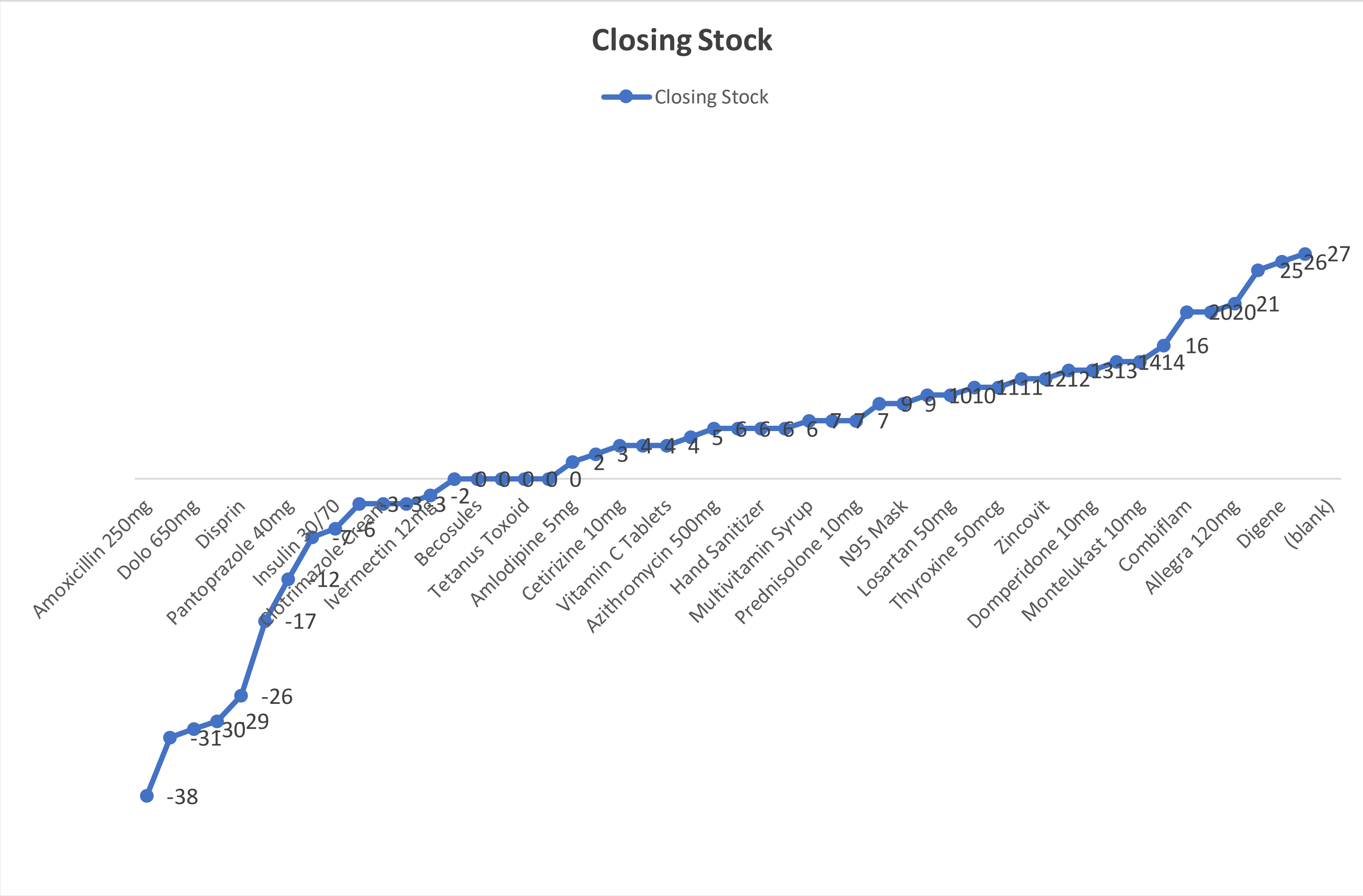
- Amoxicillin 250mg (-38 units): Critical shortage of this essential antibiotic suggests underestimation of demand, particularly during infection seasons
- ORS Sachet (-31 units): Stockout during summer months when dehydration cases peak indicates poor seasonal planning
- Dolo 650mg (-30 units): High demand for fever medication not met, potentially affecting customer loyalty

Zero Closing Stock (Perfect Balance): Products with zero closing stock indicate optimal inventory management:

- Atorvastatin 10mg, Becosules, Ibuprofen 400mg: These items show excellent demand forecasting and procurement planning

Positive Closing Stock (Overstocking): Excess inventory ties up working capital and increases expiry risk:

- Combiflam (20 units excess): Suggests over-purchasing or declining demand for this combination medicine
- Avil 25mg (16 units excess): Seasonal antihistamine with lower than expected demand

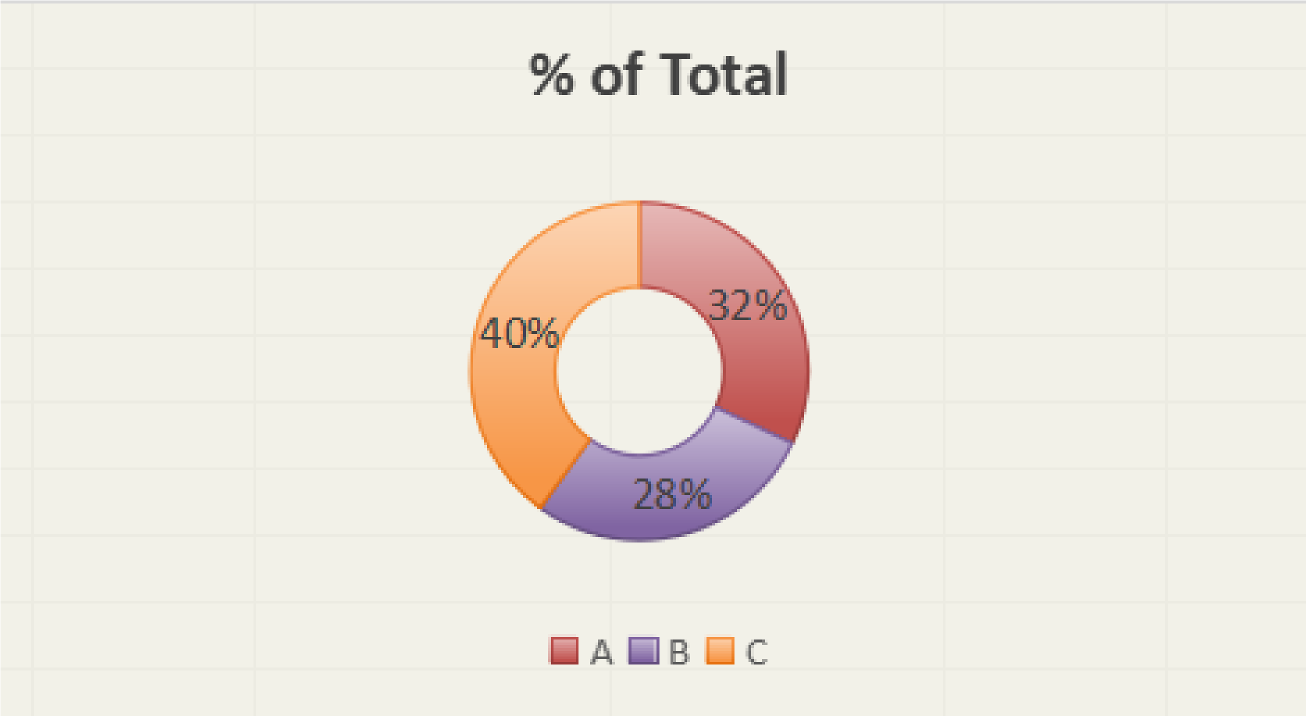


5.ABC Analysis

To strategically manage inventory, products are classified into A, B, and C categories based on their contribution to total sales revenue.

- Category A: This category contains the most valuable products that contribute the most to revenue. Items like
- Clotrimazole Cream, Volini Gel, and Allegra 120mg. These are high-priority items that require close monitoring to prevent stockouts.
- Category B: This category includes products of moderate importance to revenue, such as Azithromycin 500mg and Combiflam. These items require a moderate level of control.
- Category C: This group consists of the vast majority of products but contributes the least to revenue. Items like
- Paracetamol 500mg and N95 Mask are in this category. While they sell in high quantities, their low cost means they can be managed with less stringent inventory control policies.

Product Name	Sum of Amount (₹)	Cumulative Sales	Cumulative %	Category
Clotrimazole Cream	31157.5	31157.5	0.094736137	A
Volini Gel	25970	57127.5	0.173699388	A
Allegra 120mg	24618.96	81746.46	0.248554725	A
Multivitamin Syrup	18870	100616.46	0.305930025	A
Insulin 30/70	15834	116450.46	0.354074195	A
Hydroxychloroquine 200mg	7391.6	236992.49	0.72058904	B
Azithromycin 500mg	6253.64	243246.13	0.739603585	B
Combiflam	5459.58	248705.71	0.756203746	B
Betadine Ointment	5335	254040.71	0.772425115	B
Erythromycin 250mg	5040	259080.71	0.787749519	B
Loperamide 2mg	2242.24	309009.81	0.939561765	C
Omeprazole 20mg	2185.52	311195.33	0.946206962	C
Vitamin C Tablets	2013.66	313208.99	0.952329609	C
N95 Mask	1988	315196.99	0.958374235	C
Prednisolone 10mg	1890	317086.99	0.964120887	C
Domperidone 10mg	1861.92	318948.91	0.96978216	C
Levocetirizine 5mg	1610	320558.91	0.974677455	C
Paracetamol 500mg	1562.4	322121.31	0.979428021	C
Avil 25mg	1510.5	323631.81	0.984020781	C
Tetanus Toxoid	1305.25	324937.06	0.987989468	C



From the ABC analysis

- 32% belongs to “A” category
- 28% belongs to “B” category
- 40% belongs to “C” category

5.Data Collection & Cleaning

Not all the data was readily available in Excel format. Some parts, such as sales and purchase details, were extracted from the GST software used by the shop. This was a challenge because the software did not allow direct export into clean tables, so I had to manually copy and compile the records.

Once collected, the data was standardized — dates were formatted uniformly, product names were cleaned to remove duplicates or spelling variations, and amounts were checked for consistency. For example, certain entries had missing fields or mixed units, which I corrected manually.

This step ensured that the dataset became structured and reliable for further analysis. It also reflects the practical difficulty of working with real-world business data, where manual intervention is often required before meaningful analysis can be done.

I may have missed a few points, and this will be cleaned further before the final submission.

6.Analysis Process & Method Justification

1. Excel Pivot Tables

Since most of the sales and purchase records were maintained in tabular form, Excel pivot tables were the most practical tool for the analysis. They made it easy to quickly summarize large amounts of data, such as calculating total sales, identifying the top-selling products, and comparing sales across different months. Another advantage is that pivot tables are simple to use and can be understood even by the shop owner and staff, making the analysis more relevant and easy to replicate in daily operations.

2. Expiry Analysis

To understand the risk of stock wastage, I checked all medicines that were due to expire within the next 90 days. This is a practical threshold since most chemist shops rotate medicines within three months, and suppliers may not accept expired returns after that time. By listing products that were close to expiry, the shop can now identify which items need to be sold quickly, discounted, or reordered in smaller quantities in the future.

3.ABC Classification

The medicines were also grouped into three categories based on their contribution to overall sales:

- A class – the small set of products that bring in nearly 70% of revenue. These are critical and must always be in stock while also monitored closely for expiry.
- B class – products that contribute to the next 20% of sales. These are moderately important and require regular but less intensive tracking.
- C class – items that make up the remaining 10% of sales. These are low-value or slow-moving products, and they don't require much attention beyond ensuring basic availability.

  This was done using a simple excel formula(=IF(D9<0.7,"A",IF(D9<0.9,"B","C")))

This categorization gives the owner a clear picture of which products really matter to the business and where to focus efforts.

7.Results & Findings

- **ABC Classification**

  - Around 32% of items fall under Category A, contributing to the majority of the store's revenue. These include high-demand and high-value products such as Clotrimazole Cream, Volini Gel, Allegra 120mg, and Insulin.
  - 28% of items are Category B, which are moderately important but still need regular monitoring.

- 40% of items fall under Category C, which account for very little revenue and are mostly slow-moving medicines.
  → This clearly shows that a smaller proportion of items drive the majority of sales, highlighting where the owner should focus stocking decisions.

- **Expiry Risk Analysis**

  - Medicines with expiry dates within the next 90 days were flagged.
  - A pattern was observed where Category C items are more prone to expiry since they move slowly. This contributes directly to wastage and financial loss.
  - Without a structured system, identifying and discounting these items early becomes difficult, leading to stockpile accumulation.

- **Sales Insights from Pivot Analysis**

  - Pivot summaries helped identify top-selling products in April, with skin creams, gels, and common OTC medicines showing high demand.
  - Seasonal and lifestyle-related medicines (like allergy tablets and inhalers) also showed recurring sales patterns.
  - On the other hand, certain surgical items and less-prescribed drugs were sold in very low volumes.

- **Operational Observation**

  - Manual oversight combined with Excel tracking is not sufficient to catch expiry and stock movement trends in real time.
  - Without a structured forecasting model, stock is either over-purchased or left unsold, which directly impacts profitability.

In summary, the **findings highlight both strengths and weaknesses**. The store has a solid base of high-demand products (A category), but poor visibility into expiry and slow-moving items creates significant risks. A structured, data-driven model for tracking and forecasting can directly improve inventory rotation, reduce wastage, and increase profitability.